

Master of City Planning

SUBJECT - PLANNING INFORMATICS

“Identification of Spatial Inequality in School Accessibility Across Wards Based on Population Demand and Service Distribution - A Study of Bhubaneswar”

Submitted by

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Table of Contents

Table of Contents.....	2
1. Introduction	4
1.1. Background and Context.....	4
1.2. Problem Statement	4
1.3. Objectives.....	4
1.4. Study Area	4
2. Data Description.....	6
2.1. Datasets Used	6
2.2. Data Pre-processing in QGIS.....	6
3. Methodology	6
3.1. Geospatial Pre-processing (GeoPandas)	6
3.2. Computation of Accessibility Indicators.....	7
3.3. Correlation Analysis.....	8
3.4. K-Means Clustering	9
4. Results and Analysis	10
4.1. Summary Statistics.....	10
4.2. Spatial Distribution of Schools	11
4.3. School Accessibility per 10,000 Population	12
4.4. Demand Pressure — Population per School.....	13
4.5. Correlation Analysis Results.....	13
4.6. K-Means Cluster Analysis.....	15
Elbow Method — Optimal Number of Clusters.....	15
Cluster Classification Results	15
5. Discussion.....	17
5.1. Spatial Patterns and Inequality	17
5.2. Implications of Zero-School Wards	17
5.3. Correlation Analysis Insights	18
5.4. Limitations	18
6. Conclusion	18
7. Python Script — Complete Analysis Code	19
7.1. Library Imports and File Paths.....	19
7.2. Data Loading and Spatial Join	19
7.3. Accessibility Indicators.....	20
7.4. K-Means Clustering	20

7.5. Map Generation and Export.....	21
8. References	22

1. Introduction

1.1. Background and Context

Access to quality education is a fundamental right and a cornerstone of sustainable urban development. In rapidly growing cities across India, the spatial distribution of educational infrastructure often fails to keep pace with population growth, leading to significant disparities in school accessibility across urban wards. Bhubaneswar, the capital of Odisha and one of India's fastest-growing smart cities, presents a compelling case study for examining these spatial inequalities.

Bhubaneswar Municipal Corporation (BMC) manages 67 wards with a diverse mix of well-planned residential sectors, informal settlements, and newly urbanised peripheral areas. While the city has witnessed substantial infrastructure investment under smart city initiatives, the distribution of government school facilities remains uneven, creating educational service gaps in specific urban zones.

1.2. Problem Statement

The core urban issue addressed in this study is: How spatially unequal is government school accessibility across BMC wards when evaluated against population demand and service distribution patterns? This problem has direct implications for urban planning, resource allocation, and social equity.

Research Problem

- 27 out of 67 BMC wards (40.3%) have NO government school within their boundaries
- Mean accessibility is only 0.69 schools per 10,000 population — significantly below recommended norms
- School placement does not correlate meaningfully with population density ($r = -0.086$)
- Large portions of the population face high demand pressure with 12,401+ people per school in Very Low Accessibility wards

1.3. Objectives

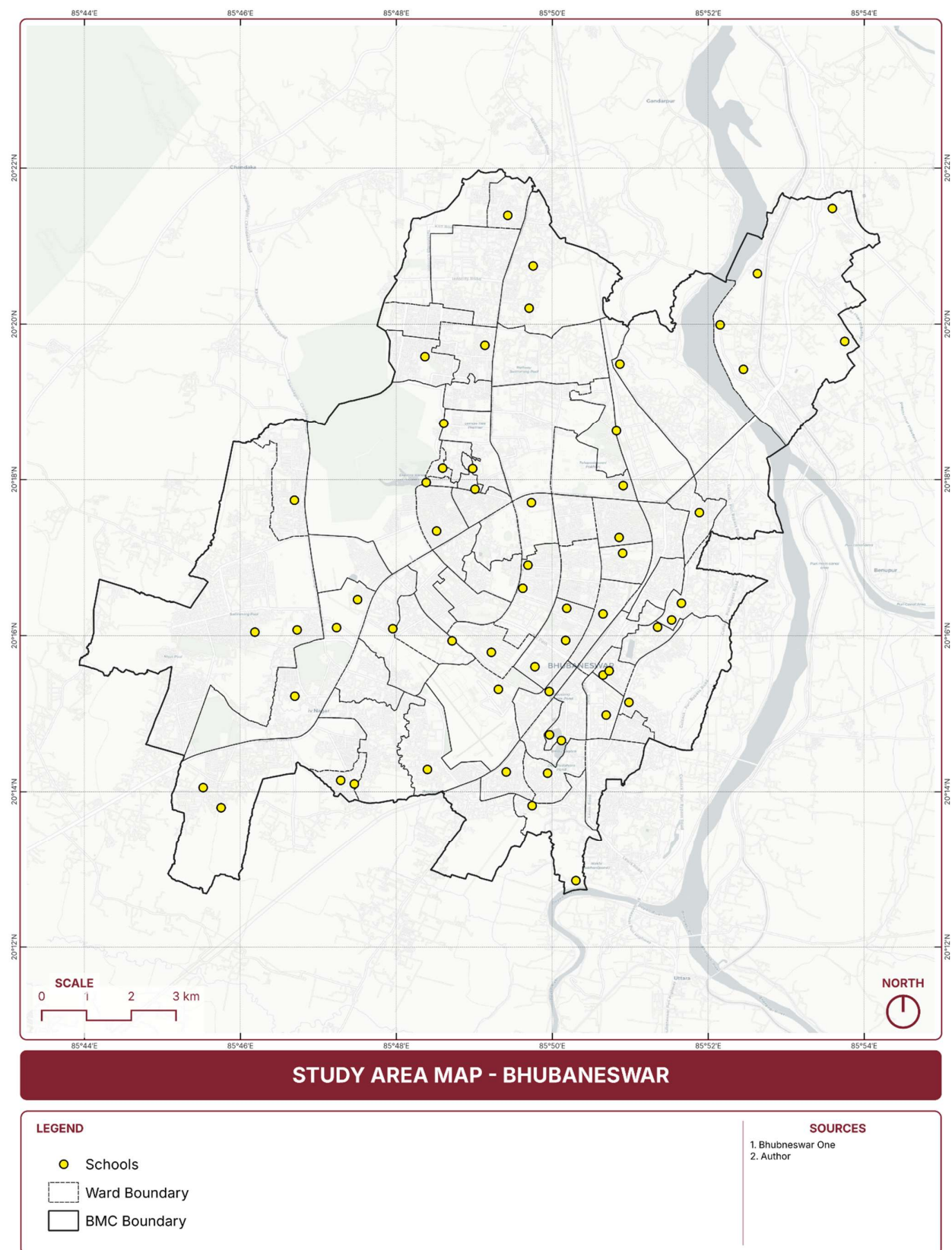
The study pursues the following specific objectives:

1. To map the spatial distribution of government schools across BMC wards using GIS-based geospatial pre-processing.
2. To compute ward-level school accessibility indicators: schools per 10,000 population, schools per sq. km, and population per school.
3. To conduct Pearson correlation analysis between population characteristics and school distribution.
4. To classify wards into accessibility categories using K-Means clustering.
5. To identify spatially underserved wards and provide data-driven planning recommendations.

1.4. Study Area

The study area encompasses all 67 wards of the Bhubaneswar Municipal Corporation (BMC), Odisha, India. Bhubaneswar is the state capital and a major smart city with a planned urban core and rapidly growing peripheral areas. The city is located in eastern India (approximately 20.3°N, 85.8°E) and covers an area of approximately 135 sq. km. The city's population of approximately

1 million is distributed unevenly across its 67 wards, making it an ideal case for spatial accessibility analysis.



Map 1 Study Area — Bhubaneswar School Locations
Source: BhubaneswarOne, Author

2. Data Description

2.1. Datasets Used

The study uses three primary spatial and attribute datasets, all sourced from official BMC records and compiled in GeoPackage (.gpkg) format:

Table 1 Primary Datasets Used for Spatial Analysis

Dataset		Type	Attributes		Source
Ward (BMC)	Boundary	Polygon Shapefile (.gpkg)	wardno, Pop_Dens_1, Shape_Area	totalwardp, Area_km2_	BhubaneswarOne, Census 2011
Government Schools		Point Shapefile (.gpkg)	School Nam (School Name)		BhubaneswarOne
Ward Population & Density	Attribute (embedded in Ward Boundary)		Population, Pop. Density (persons/km ²)		Census 2011 / BMC Records

2.2. Data Pre-processing in QGIS

Both spatial layers were first loaded into QGIS 3.44.2 and reprojected to EPSG:32645 (WGS 84 / UTM Zone 45N), the appropriate projected coordinate reference system (CRS) for Bhubaneswar. This CRS ensures that all distance and area calculations are performed in metric units (metres), which is essential for computing meaningful accessibility indicators.

A spatial join (Join Attributes by Location — Summary) was performed in QGIS to count the number of school points falling within each ward polygon, adding a school_count attribute to the ward layer. The processed ward layer was then exported as a CSV and GeoPackage for use in the Python analysis pipeline.

3. Methodology

The analytical framework follows a four-stage geospatial and statistical pipeline, as Data Loading → Accessibility Indicators → Correlation → K-Means Clustering

3.1. Geospatial Pre-processing (GeoPandas)

Both spatial layers were loaded using GeoPandas, reprojected to EPSG:32645, and spatially joined to assign school counts to each ward. Area was computed directly from ward geometry (sq. km) to ensure consistency independent of pre-existing attribute field names.

```
import geopandas as gpd
```

```

# Load spatial layers

wards = gpd.read_file('Ward boundary.gpkg')
schools = gpd.read_file('Govt Schools.gpkg')

# Reproject to projected CRS (EPSG:32645 — UTM Zone 45N for Bhubaneswar)
wards = wards.to_crs(epsg=32645)
schools = schools.to_crs(epsg=32645)

# Spatial join — count schools per ward
schools_in_wards = gpd.sjoin(schools, wards[['wardno', 'geometry']],
                             how='left', predicate='within')

school_count = schools_in_wards.groupby('wardno').size().reset_index(name='school_count')
wards = wards.merge(school_count, on='wardno', how='left')
wards['school_count'] = wards['school_count'].fillna(0).astype(int)

# Compute area from geometry (sq km)
wards['area_km2'] = wards.geometry.area / 1_000_000

```

3.2. Computation of Accessibility Indicators

Three ward-level accessibility indicators were derived to capture different dimensions of educational service provision:

Table 2 Ward-Level School Accessibility Indicators — Formula and Interpretation

Indicator	Formula	Interpretation
Schools per 10,000 Population	$(\text{school_count} / \text{population}) \times 10,000$	Relative provision per unit of population
Schools per sq. km	$\text{school_count} / \text{area_km2}$	Spatial density of school coverage
Population per School	$\text{population} / \text{school_count}$	Demand pressure on each school

For wards with zero schools, the population per school indicator is assigned the full ward population to represent the theoretical demand burden in the absence of any school facility.

```
# Rename columns to standardized names

wards = wards.rename(columns={'totalwardp': 'population',
                              'POP_Dens_1': 'pop_density',
                              'Area_km2_': 'area_km2'})

# Accessibility Indicator 1: Schools per 10,000 population

wards['schools_per_10k'] = (wards['school_count'] / wards['population']) * 10000

wards['schools_per_10k'] = wards['schools_per_10k'].replace([np.inf, -np.inf],
0).fillna(0)

# Accessibility Indicator 2: Schools per sq km

wards['schools_per_sqkm'] = wards['school_count'] / wards['area_km2']

wards['schools_per_sqkm'] = wards['schools_per_sqkm'].replace([np.inf, -np.inf],
0).fillna(0)

# Accessibility Indicator 3: Population per school (demand pressure)

wards['pop_per_school'] = wards['population'] / wards['school_count'].replace(0,
np.nan)

wards['pop_per_school'] = wards['pop_per_school'].fillna(wards['population'])
```

3.3. Correlation Analysis

Pearson correlation analysis was conducted between all computed indicators and ward characteristics (population, density, area). This revealed the strength and direction of relationships between school distribution and demographic variables, helping to identify whether population pressure drives school placement or if systemic gaps exist.

```
from scipy.stats import pearsonr

corr_vars = ['school_count', 'schools_per_10k', 'schools_per_sqkm',
             'pop_per_school', 'population', 'pop_density', 'area_km2']
```



```
# Correlation matrix
corr_matrix = wards[corr_vars].corr()

# Key Pearson correlations with p-values
r1, p1 = pearsonr(wards['population'], wards['school_count'])
r2, p2 = pearsonr(wards['pop_density'], wards['schools_per_10k'])
# Results: r=0.186 (p=0.1309) and r=-0.086 (p=0.4869)
```

3.4. K-Means Clustering

K-Means clustering was applied to classify wards into four distinct accessibility groups based on the multi-dimensional accessibility indicator profile. The four input features were standardised using StandardScaler before clustering to ensure equal weighting. The optimal number of clusters (k=4) was determined using the Elbow Method.

```
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

# Features for clustering
cluster_features = ['schools_per_10k', 'schools_per_sqkm',
                    'pop_per_school', 'pop_density']

# Standardise features
X = wards[cluster_features].copy()
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Apply K-Means with k=4
kmeans = KMeans(n_clusters=4, random_state=42, n_init=10)
wards['cluster'] = kmeans.fit_predict(X_scaled)

# Label clusters by ascending schools_per_10k mean
cluster_means = wards.groupby('cluster')['schools_per_10k'].mean().sort_values()
```

```

labels = ['Very Low Accessibility', 'Low Accessibility',
          'Moderate Accessibility', 'High Accessibility']
cluster_labels = {cid: lbl for (cid, _), lbl in zip(cluster_means.items(), labels)}
wards['accessibility_class'] = wards['cluster'].map(cluster_labels)

```

4. Results and Analysis

4.1. Summary Statistics

The analysis of 67 BMC wards with 58 government schools revealed the following key statistics:

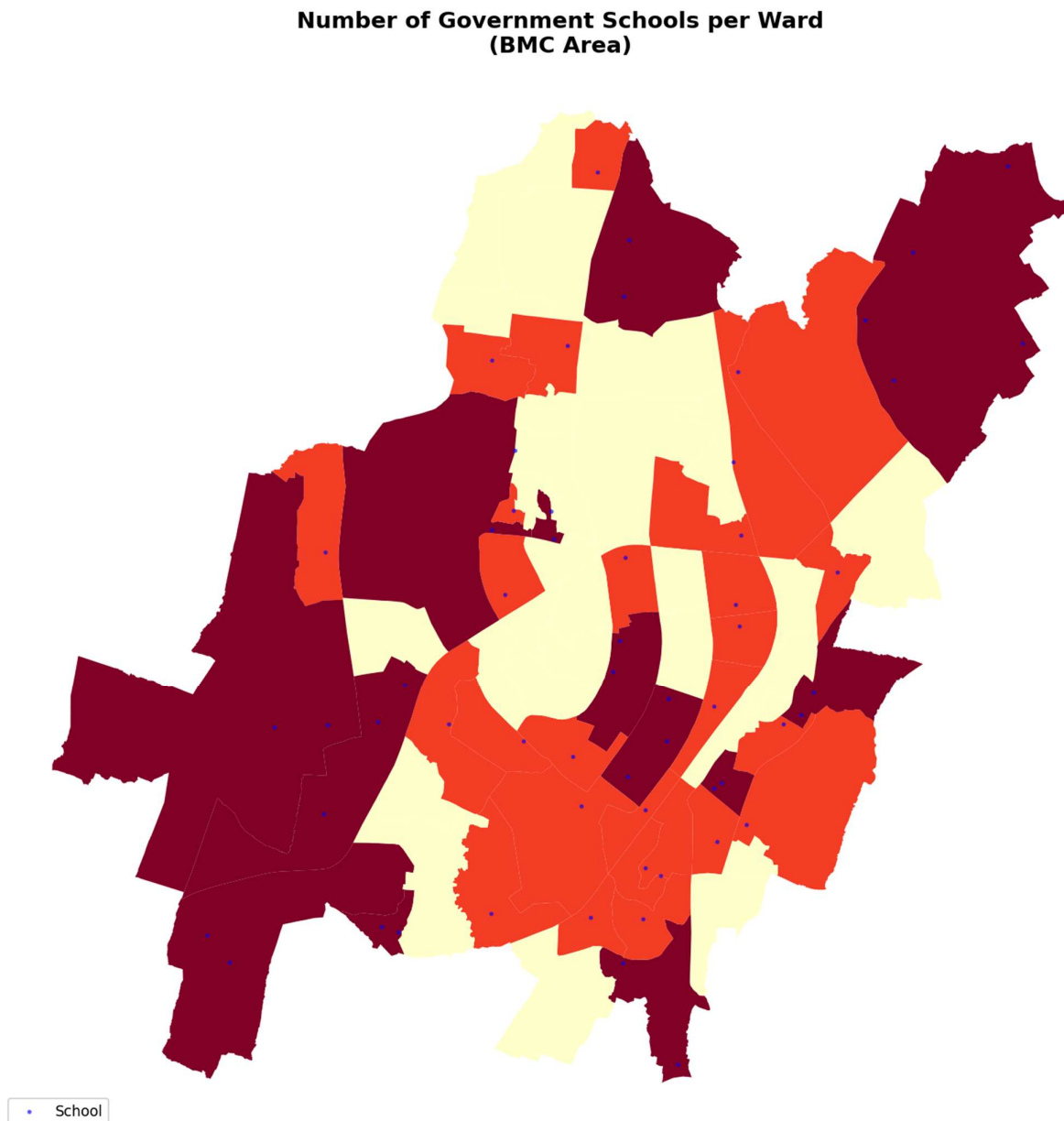
Table 3 Summary Statistics — Government School Distribution across 67 BMC Wards

Statistic	Value
Total Wards Analyzed	67
Total Government Schools	58
Average Schools per Ward	0.9
Min Schools in a Ward	0
Max Schools in a Ward	5
Wards with Zero Schools	27 (40.3%)
Min Schools per 10k Population	0.00
Max Schools per 10k Population	3.09
Mean Schools per 10k Population	0.69
Average Population per School	11,216

These statistics immediately highlight the severity of school accessibility inequality in Bhubaneswar. With an average of less than one school per ward and 40% of wards having no government school at all, the city faces a significant challenge in ensuring equitable educational access.

4.2. Spatial Distribution of Schools

Map 2 shows the number of government schools per ward across BMC. A clear spatial clustering of schools is visible in the central and older parts of the city, while peripheral wards — particularly those with recent urban growth — show few or no schools.



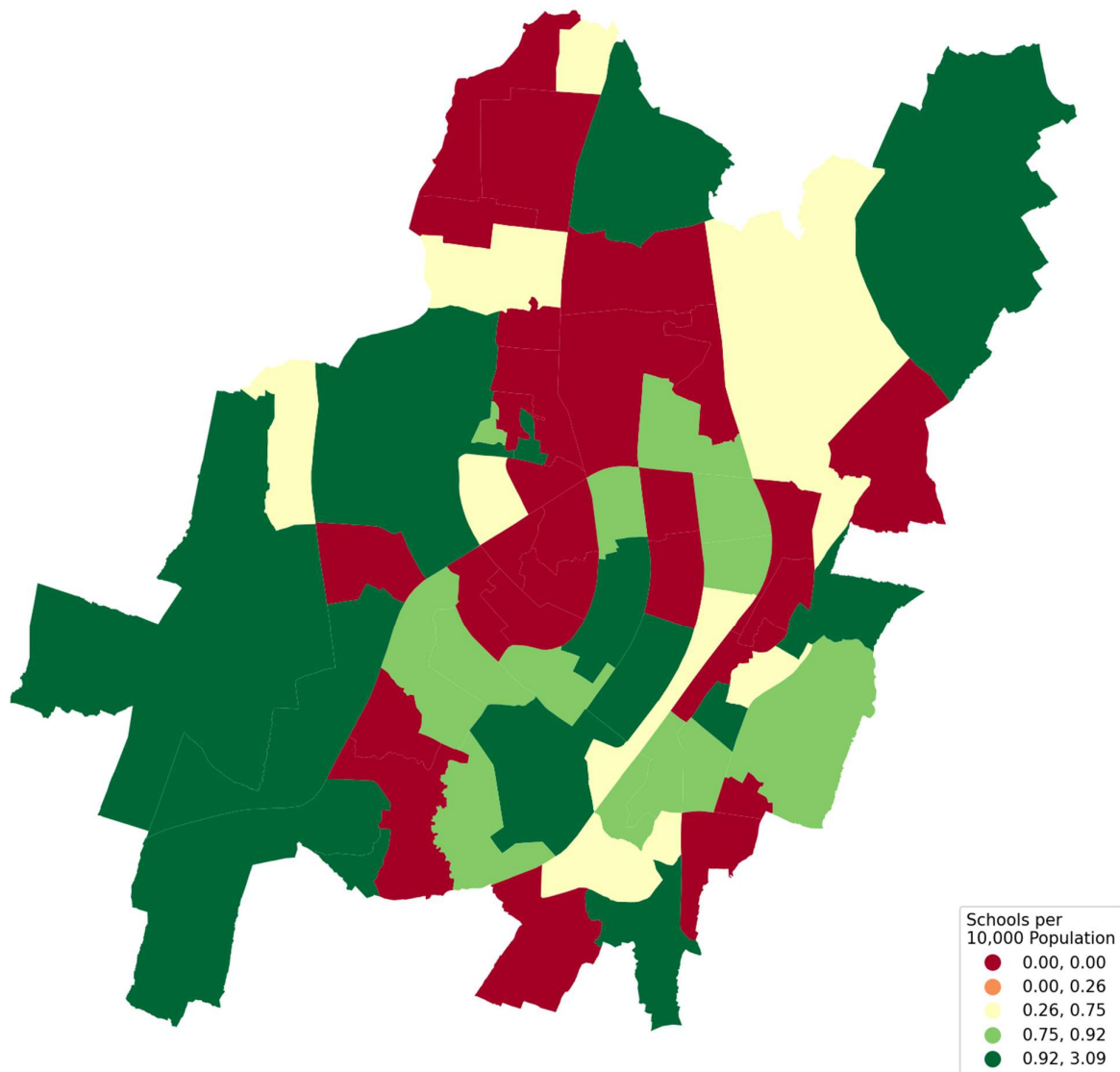
Map 2 Spatial Distribution of Government Schools per BMC Ward (Choropleth Map)

Source: BhubaneswarOne, Author

4.3. School Accessibility per 10,000 Population

Map 3 presents the ward-level accessibility indicator: schools per 10,000 population. This normalised indicator reveals that even wards with a small number of schools may have relatively better accessibility if their population is low. Conversely, densely populated wards with only 1–2 schools show very poor per-capita accessibility.

School Accessibility: Schools per 10,000 Population per Ward



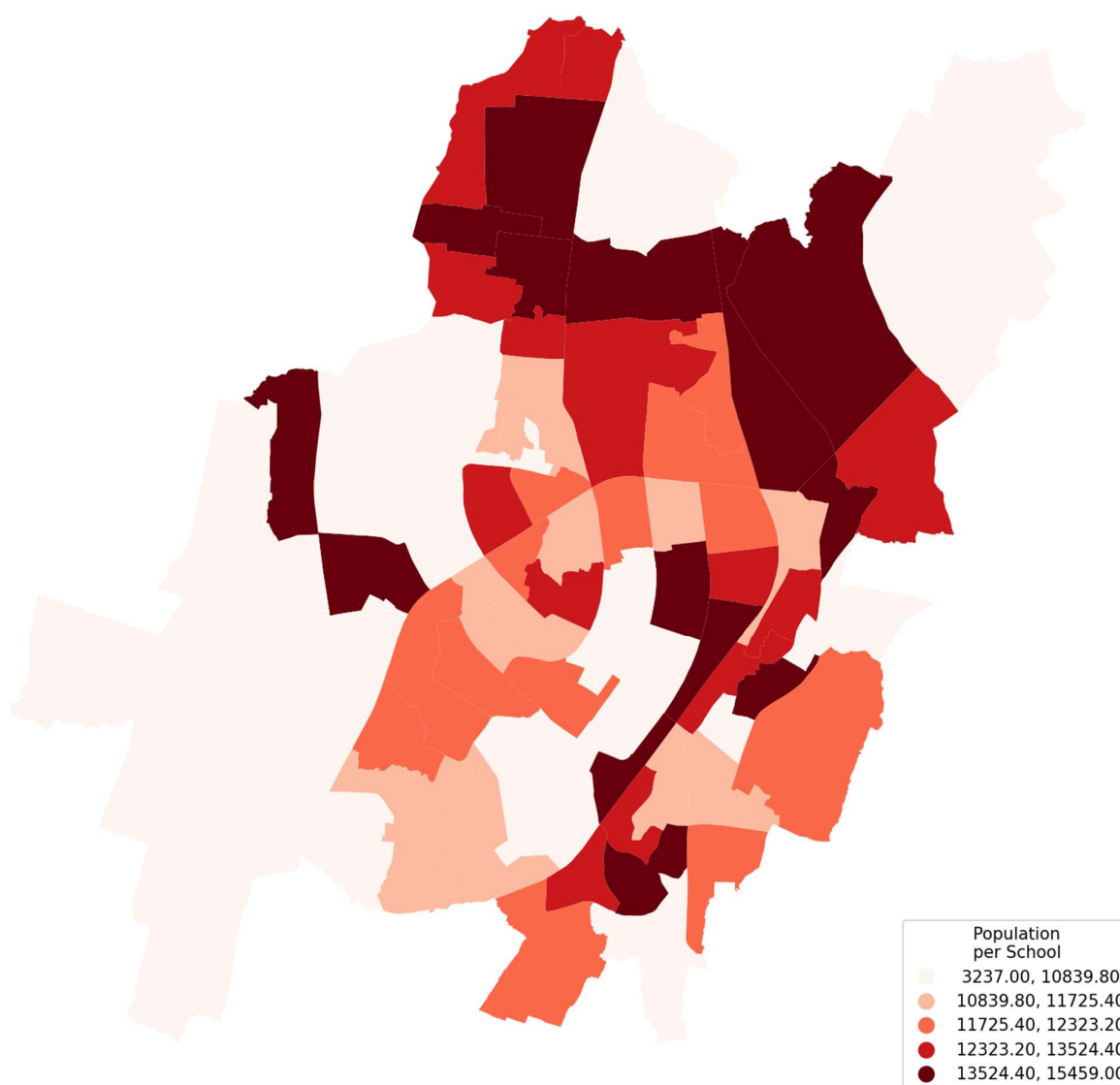
Map 3 School Accessibility per 10,000 Population across BMC Wards

Source: BhubaneswarOne, Author

4.4. Demand Pressure — Population per School

Map 4 maps the population per school indicator, which represents the demand burden on each existing school. Wards with zero schools are assigned the full ward population as the demand burden, making this map a particularly stark visual of educational service gaps. Wards showing population-per-school values above 12,000 are considered critically underserved by international norms.

Demand Pressure: Population per School per Ward



Map 4 Demand Pressure — Population per School per BMC Ward

Source: BhubaneswarOne, Author

4.5. Correlation Analysis Results

The Pearson correlation analysis produced the following key results:

Table 4 Pearson Correlation Analysis Results — School Accessibility and Ward Characteristics

Variable Pair	r Value	p-Value	Interpretation
Population vs. School Count	0.186	0.131	Weak positive — not statistically significant ($p > 0.05$)
Pop. Density vs. Schools per 10k	-0.086	0.487	Very weak negative — not statistically significant

The correlation analysis reveals a critically important finding: school placement in BMC wards shows no statistically significant relationship with either population size or population density. This confirms that school distribution is not demand-driven and is likely influenced by historical patterns, land availability, and ad hoc planning decisions rather than systematic service area planning.

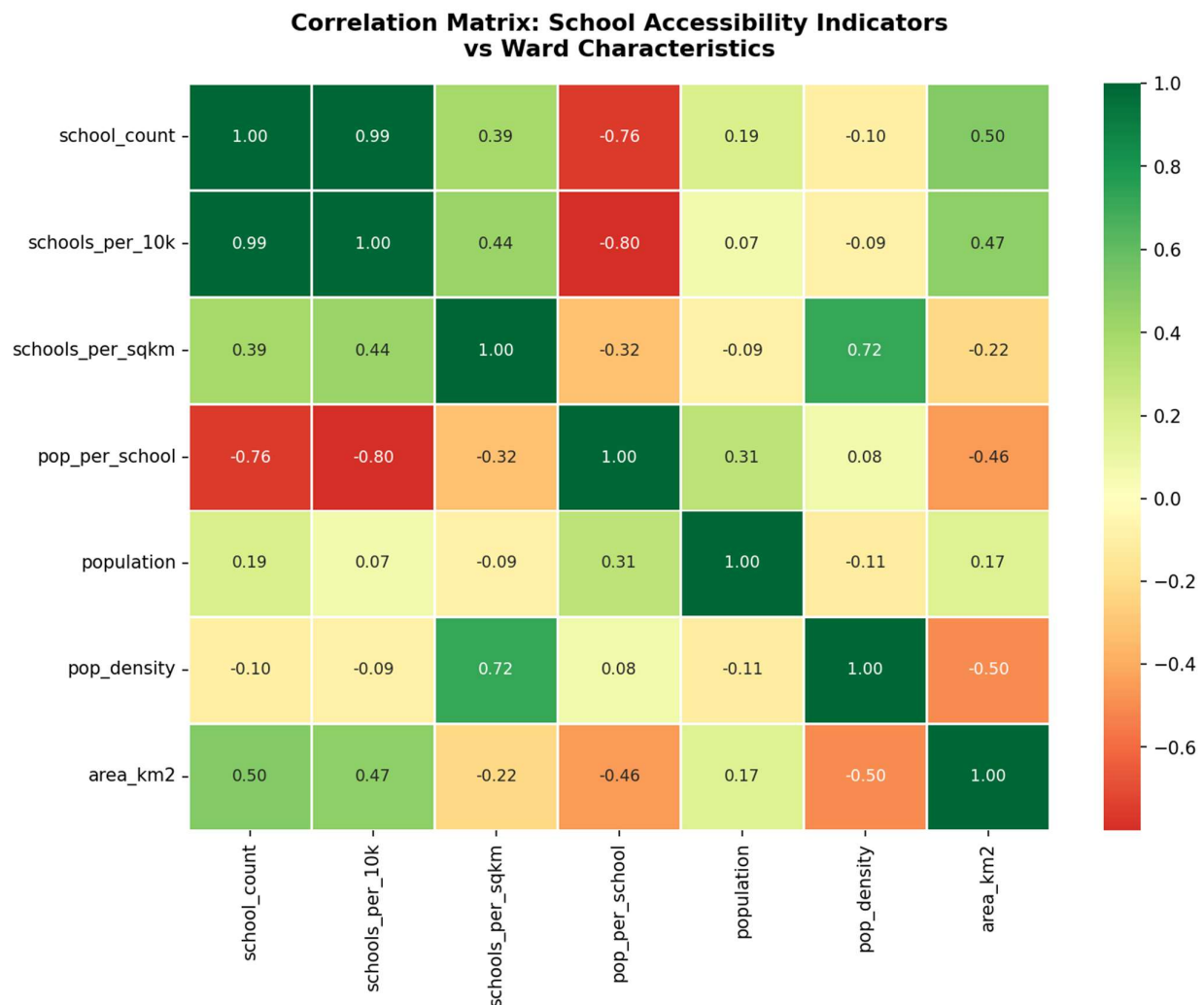


Figure 1 Pearson Correlation Matrix of School Accessibility Indicators and Ward Characteristics

Source: Author

4.6. K-Means Cluster Analysis

Elbow Method — Optimal Number of Clusters

The Elbow Method was applied across $k = 2$ to 7 clusters. The within-cluster sum of squares (inertia) showed a clear elbow at $k = 4$, confirming that four clusters provide the optimal balance between model complexity and explanatory power.

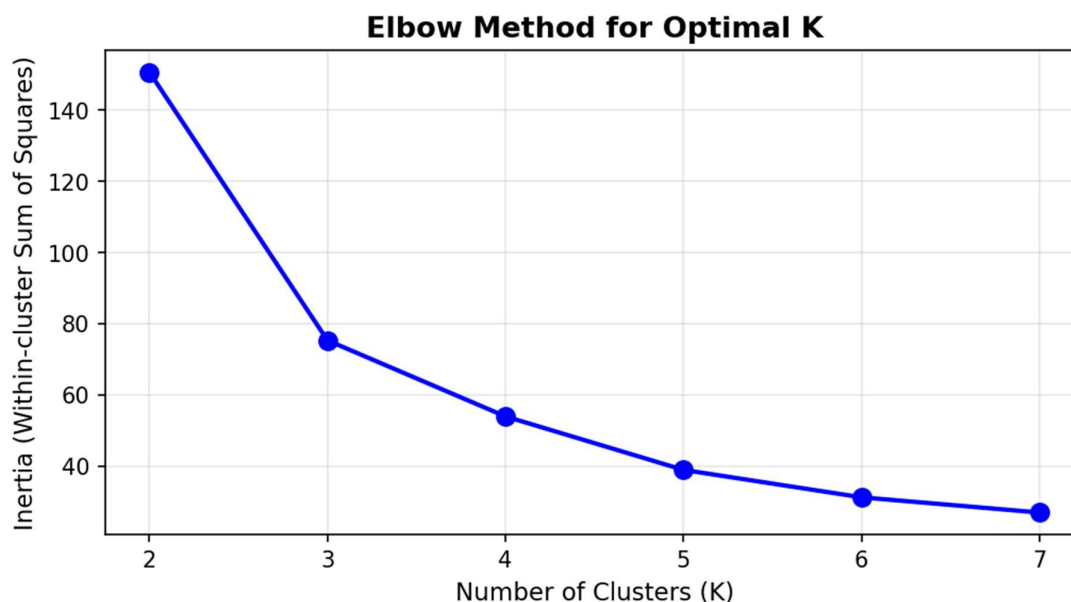


Figure 2 Elbow Method — Optimal Number of Clusters ($k = 4$ selected)

Source: Author

Cluster Classification Results

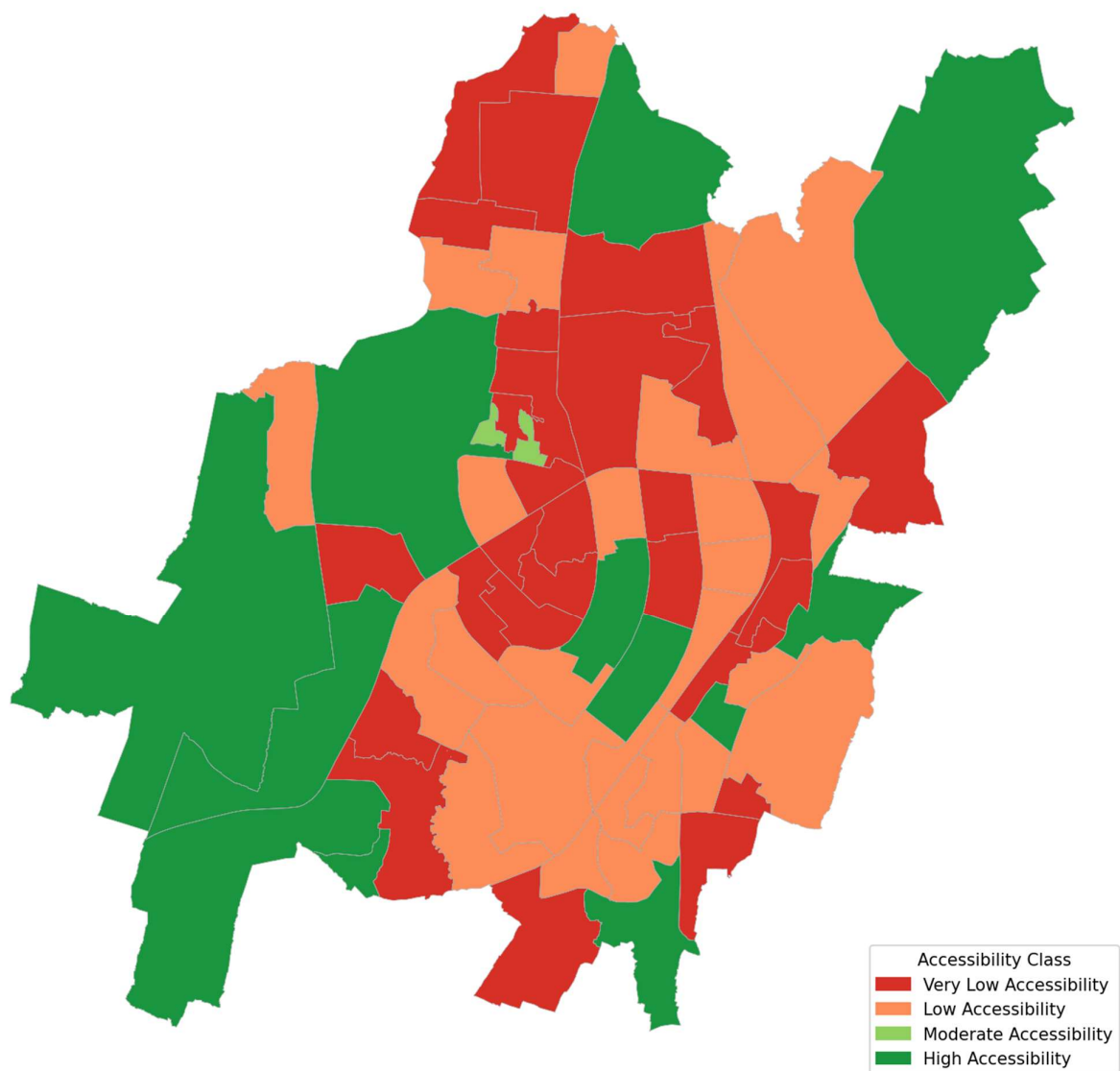
The K-Means algorithm classified all 67 wards into four distinct accessibility categories. The cluster distribution and mean indicator values are presented below:

Table 5 K-Means Cluster Classification — Ward Distribution and Mean Accessibility Indicators

Accessibility Class		Ward Count		Avg. Schools/Ward	Schools per 10k Pop	Pop. per School
High Accessibility		12	17.9%	2.42	1.92	5,412
Moderate Accessibility		2	3.0%	1.50	1.26	8,702
Low Accessibility		26	38.8%	1.00	0.78	12,858
Very Low Accessibility	Low	27	40.3%	0.00	0.00	12,401

The results reveal a highly skewed distribution where the vast majority of wards (79.1%) fall in the Very Low or Low Accessibility categories. Only 12 wards (17.9%) achieve High Accessibility, while a mere 2 wards are classified as Moderate. This distribution underscores the severity and breadth of spatial inequality in school access across Bhubaneswar.

**K-Means Clustering: Ward Classification
by School Accessibility Patterns**



Map 5 K-Means Cluster Map — Ward Classification by School Accessibility Pattern

Source: Author

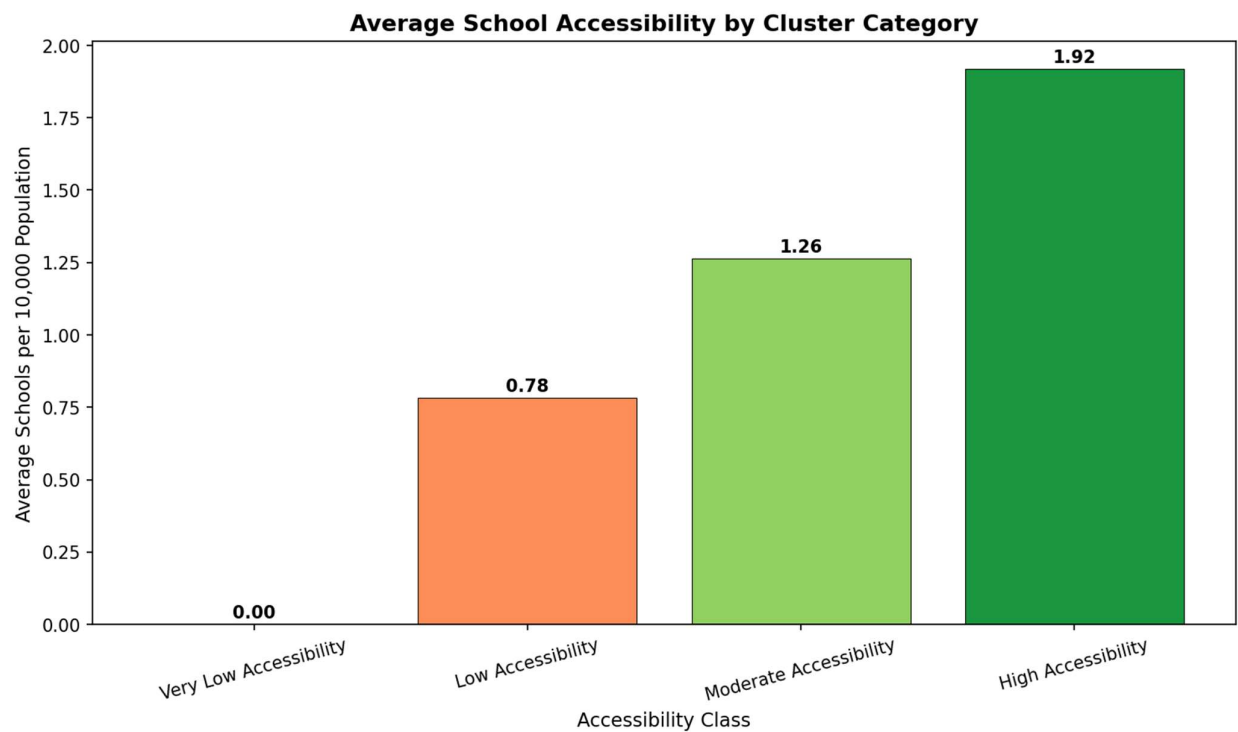


Figure 3 Average School Accessibility by Cluster Category
Source: Author

5. Discussion

5.1. Spatial Patterns and Inequality

The spatial analysis reveals a clear centre-periphery gradient in school accessibility across Bhubaneswar. The 12 High Accessibility wards are concentrated in the older, planned sectors of the city (notably the central BDA-planned areas and established neighbourhoods), while Very Low Accessibility wards form a ring around the periphery where rapid urban expansion has outpaced infrastructure provision.

This pattern is characteristic of Indian cities where government school networks were historically established in the urban core and have not been expanded to match the growth of peripheral settlements. The very weak correlation between population density and school accessibility ($r = -0.086$) further suggests that current distribution patterns do not follow principles of service area planning or population-based allocation.

5.2. Implications of Zero-School Wards

The finding that 27 wards (40.3%) have zero government schools is alarming from an equity planning perspective. Children in these wards must travel to adjacent wards for government schooling, which creates barriers of distance, transport cost, and safety — disproportionately affecting lower-income families who depend on government schools. The average population-per-school burden of 12,401 in Very Low Accessibility wards (compared to 5,412 in High Accessibility wards) implies that existing schools in better-served areas may also be over-enrolled and under-resourced.

5.3. Correlation Analysis Insights

The statistically insignificant correlation between population and school count ($r = 0.186$, $p = 0.131$) is a particularly important finding. In a demand-responsive planning system, one would expect a strong positive correlation — more people should lead to more schools. The absence of this relationship suggests that school locations in BMC wards are the result of historical legacy, land availability, or political factors rather than systematic spatial planning based on population need.

5.4. Limitations

The study acknowledges the following limitations:

- Only government schools are included; private schools, which may compensate for government supply gaps in some wards, are not analysed.
- Student catchment data and actual enrolment figures are unavailable, limiting the ability to assess actual versus theoretical demand.
- The analysis uses ward-level aggregates; intra-ward spatial variation in school accessibility is not captured.

6. Conclusion

This study provides comprehensive empirical evidence of significant spatial inequality in government school accessibility across the 67 wards of Bhubaneswar Municipal Corporation. Using an integrated geospatial and statistical methodology — combining GeoPandas preprocessing, accessibility indicator computation, Pearson correlation analysis, and K-Means clustering — the study produces a multi-layered spatial understanding of educational service distribution.

The key finding that 79.1% of BMC wards fall in Very Low or Low Accessibility categories, combined with the statistically insignificant correlation between population and school distribution, reveals a systemic planning failure. School locations in Bhubaneswar appear to reflect historical patterns rather than contemporary population needs or equity principles.

The spatial outputs generated by this study — including four choropleth maps, a correlation matrix, a cluster classification map, and a ward-level accessibility dataset — provide actionable intelligence for BMC planners, urban policy makers, and educational administrators. The cluster-based classification of wards into four accessibility tiers offers a practical planning tool for prioritising interventions and tracking progress toward equitable school distribution.

This study demonstrates that geospatial informatics, when applied to publicly available urban datasets, can generate powerful insights to support evidence-based planning for social infrastructure equity — a critical need in India's rapidly urbanising cities.

7. Python Script — Complete Analysis Code

The complete Python script used for this analysis is provided below. The script covers all steps from data loading to map generation and output export.

7.1. Library Imports and File Paths

```
import geopandas as gpd
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import seaborn as sns

from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from scipy.stats import pearsonr

import warnings, os
warnings.filterwarnings('ignore')

# File Paths (update to your local paths)
WARD_SHP = r'S:\01 MCP\SEM 2\PLANNING INFORMATICS\01 project\Ward boundary.gpkg'

SCHOOLS_SHP = r'S:\01 MCP\SEM 2\PLANNING INFORMATICS\01 project\Govt Schools.gpkg'

OUTPUT_FOLDER = r'S:\01 MCP\SEM 2\PLANNING INFORMATICS\01 project\outputs\'

os.makedirs(OUTPUT_FOLDER, exist_ok=True)
```

7.2. Data Loading and Spatial Join

```
# Load and reproject spatial data
wards = gpd.read_file(WARD_SHP).to_crs(epsg=32645)
schools = gpd.read_file(SCHOOLS_SHP).to_crs(epsg=32645)
```

```
# Spatial join — count schools per ward

schools_in_wards = gpd.sjoin(schools, wards[['wardno','geometry']],
                             how='left', predicate='within')

school_count = schools_in_wards.groupby('wardno').size().reset_index(name='school_count')

wards = wards.merge(school_count, on='wardno', how='left')

wards['school_count'] = wards['school_count'].fillna(0).astype(int)

print(f'Wards: {len(wards)}, Schools mapped: {wards["school_count"].sum()}')
```

7.3. Accessibility Indicators

```
# Rename columns

wards = wards.rename(columns={'totalwardp':'population',
                             'POP_Dens_1':'pop_density', 'Area_km2':'area_km2'})

wards['area_km2'] = wards.geometry.area / 1_000_000 # compute from geometry

wards['schools_per_10k'] = (wards['school_count'] / wards['population']) * 10000
wards['schools_per_sqkm'] = wards['school_count'] / wards['area_km2']
wards['pop_per_school'] = wards['population'] / wards['school_count'].replace(0, np.nan)

wards['pop_per_school'] = wards['pop_per_school'].fillna(wards['population'])

# Replace inf values
for col in ['schools_per_10k','schools_per_sqkm']:
    wards[col] = wards[col].replace([np.inf, -np.inf], 0).fillna(0)
```

7.4. K-Means Clustering

```
cluster_features = ['schools_per_10k','schools_per_sqkm','pop_per_school','pop_density']
X_scaled = StandardScaler().fit_transform(wards[cluster_features])

kmeans = KMeans(n_clusters=4, random_state=42, n_init=10)
```

```
wards['cluster'] = kmeans.fit_predict(X_scaled)

cluster_means = wards.groupby('cluster')['schools_per_10k'].mean().sort_values()
labels = ['Very Low Accessibility', 'Low Accessibility',
          'Moderate Accessibility', 'High Accessibility']
cluster_labels = {cid: lbl for (cid, _), lbl in zip(cluster_means.items(), labels)}
wards['accessibility_class'] = wards['cluster'].map(cluster_labels)

print(wards['accessibility_class'].value_counts())
```

7.5. Map Generation and Export

```
color_map = {'Very Low Accessibility': '#d73027', 'Low Accessibility': '#fc8d59',
             'Moderate Accessibility': '#91cf60', 'High Accessibility': '#1a9641'}

fig, ax = plt.subplots(figsize=(12,10))
for cls, color in color_map.items():
    wards[wards['accessibility_class']==cls].plot(ax=ax, color=color, label=cls)
wards.boundary.plot(ax=ax, color='gray', linewidth=0.4)
ax.set_title('K-Means Clustering: Ward Classification by School Accessibility')
ax.set_axis_off()

patches = [mpatches.Patch(color=c, label=l) for l,c in color_map.items()]
ax.legend(handles=patches, loc='lower right', title='Accessibility Class')
plt.savefig(OUTPUT_FOLDER + 'map_kmeans_clusters.png', dpi=150, bbox_inches='tight')

# Export final GeoPackage for QGIS import
wards.to_file(OUTPUT_FOLDER + 'ward_accessibility_final.gpkg', driver='GPKG')
```

8. References

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